

CHD Rapid-Review Cheat Sheet (Module 5)

C1 - Circulation & cyanosis (the framework)

- **Fetal:** placenta oxygenates; **umbilical VEIN carries the most-oxygenated blood** (Q12). Right-to-left shunts (foramen ovale, duct) bypass the lungs. At birth the first breath drops PVR, the duct and foramen close.
- **Cyanosis** = ≥ 3 to 5 g/dL of desaturated Hb (visible sooner if polycythaemic, later if anaemic). Central cyanosis + no response to O₂ = a right-to-left shunt.
- **Duct-dependent lesions:** keep the duct open with **PGE1**. Duct-dependent PULMONARY (cyanotic, decreased flow: pulmonary/tricuspid atresia, severe TOF) and duct-dependent SYSTEMIC (critical coarctation, HLHS). **Marked cyanosis + decreased pulmonary vascularity = PGE1** (Q6).
- **Sort every lesion by lung blood flow:** cyanotic + DECREASED flow (C2), cyanotic + INCREASED flow (C3), acyanotic left-to-right shunt (C4).
- **PVR** = (mean PA pressure - mean LA pressure) / pulmonary blood flow Q_p = transpulmonary gradient / Q_p; Wood units (normal ~1-3, >3 raised). Q20-2018 written answer.

C2 - Cyanotic, DECREASED pulmonary flow

- **Tetralogy of Fallot:** VSD + overriding aorta + RV outflow obstruction + RVH; **boot-shaped heart, RAD**. Q19: cyanosis worse with crying + RAD = TOF (RAD distinguishes it from tricuspid atresia's LAD).
- **Tet spell:** RVOT spasm to more right-to-left shunt to deep cyanosis. Treat: knees-to-chest (raise SVR), O₂, morphine, fluids, beta-blocker, phenylephrine. **NOT digoxin** (a positive inotrope worsens dynamic RVOT obstruction) = Q8 answer D.
- **Tricuspid atresia:** no RV inflow, small RV, needs an ASD + VSD to survive; **LAD. Pulmonary atresia:** duct-dependent pulmonary, PGE1.

C3 - Cyanotic, INCREASED pulmonary flow

- **D-TGA:** two parallel circuits, deeply cyanotic newborn; needs mixing (PGE1 + balloon atrial septostomy), then arterial switch. **TAPVR:** "snowman", obstructed = emergency. **Truncus:** one great artery + VSD. **DORV**.
- Q5: cyanotic + increased flow EXCEPT **D-TGA with VSD + severe PS** (that has DECREASED flow). Q18: side-by-side great arteries + increased markings + loud S₂, no murmur = **DORV (subaortic VSD, no PS)**.

C4 - Acyanotic left-to-right shunts

- **ASD:** **fixed, widely split S₂**. Secundum = RAD + RSR' in V1 (Q3, Q4-axis); primum (AVSD spectrum) = LAD. 52 yo, SOB, RAD + RVH, dilated RV = secundum ASD (Q3).
- **VSD:** harsh **holosystolic murmur at the LLSB**. Close if symptomatic / large / high Q_p:Q_s / raised PA pressure; a 100 mmHg gradient across the VSD = small restrictive (NOT an indication) = Q9 answer C.
- **PDA:** continuous **machinery murmur**. Q4: not all silent ducts must be closed (answer C). **AVSD (complete):** trisomy 21, poor feeding, tachypnoea, LLSB murmur, cardiomegaly, LAD (Q7 answer B).

C5 - Obstructive & other

- **Coarctation:** narrowing at the isthmus; **upper-limb hypertension + weak, delayed femorals (radio-femoral delay)**, murmur over the back, rib notching. Assoc: **bicuspid aortic valve, Turner**. Critical neonate = duct-dependent systemic (PGE1). Preductal (infantile) vs postductal (adult).

- **Valvar pulmonary stenosis:** ejection click + murmur at the **upper LEFT sternal border**, RVH, balloon valvuloplasty. Q13: Noonan = **pulmonary** stenosis (dysplastic valve), NOT mitral (answer E).
- **Ebstein:** tricuspid leaflets displaced apically, atrialized RV, severe TR, **wall-to-wall heart**; assoc **ASD/PFO, WPW/pre-excitation, widely split S1, maternal lithium**. Q16: key says A (ASD), but ASD is strongly associated, so the defensible EXCEPT is **B, paradoxical S2 splitting** (Ebstein gives a WIDELY split S2). Confirm with course.
- **ccTGA (L-TGA):** double discordance, physiologically corrected (pink), but the morphologic RV is systemic (fails late) + high rate of complete heart block. **Vascular ring:** stridor + feeding difficulty; work-up = barium swallow + CT/MRI, **echo is LIMITED** (Q11 answer D).
- **Isomerism:** RIGHT isomerism = two right sides = **ASPLENIA** (Howell-Jolly, severe cyanotic CHD); LEFT = two left sides = polysplenia + interrupted IVC + heart block. Q15: key says B (polysplenia) but standard teaching = ASPLENIA; confirm with course.

C6 - Single-ventricle pathway (Glenn & Fontan)

- One usable ventricle: it pumps the BODY, and the systemic veins are plumbed onto the PA to flow through the lungs **PASSIVELY (no pump)**. So the whole pathway needs a **LOW PVR**.
- **Stage 1 is tailored:** obstructed aorta (HLHS) = **Norwood** (rebuild the main PA into a neo-aorta off the RV + a small controlled BT or Sano shunt + atrial septectomy); too MUCH lung flow = **PA band**; too LITTLE = **BT shunt**. All then go Glenn then Fontan.
- **Bidirectional Glenn:** SVC to PA (upper-body blood passive to the lungs); shunt taken down; IVC still to the heart (mild cyanosis). **Fontan:** IVC also to PA (total cavopulmonary) + a fenestration; ventricle pumps only the body, acyanotic.
- **Q2 (Glenn prerequisites EXCEPT):** answer **B** - a HIGH PVR (>3 Wood units) is a CONTRAINDICATION, not a prerequisite. Need low PVR, low PA pressure, good ventricle, good pulmonary arteries, competent AV valve.
- **Q10 (Glenn complications EXCEPT):** answer **E, protein-losing enteropathy** - that is a FONTAN complication (needs the chronically high whole-body venous pressure). Glenn only raises the SVC.
- **Mechanisms:** Glenn to **pulmonary AVMs** = the lungs get only SVC blood, so the liver's "hepatic factor" (in IVC blood) never reaches them (the Fontan cures it). Every **Fontan complication** = high whole-body venous pressure (lymph leaks: PLE into gut, plastic bronchitis into airway; liver congestion) + slow stagnant flow (clots) + no pump for the lungs (exercise intolerance, Fontan failure); atrial suture lines give the arrhythmias.

C7 - Adult CHD (the late lesion the repair left behind)

- **Repaired TOF:** the transannular patch cures the obstruction but leaves free **pulmonary regurgitation** to RV dilation to dysfunction to arrhythmia / sudden death; treat with a **timed pulmonary valve replacement**. Q1: EARLIER repair is better, so "the later the repair the better" is the FALSE statement (answer A).
- **Adult D-TGA (by operation):** atrial switch (Mustard/Senning) leaves the morphologic RV as the systemic ventricle (fails late; baffle leak/obstruction; atrial arrhythmia); arterial switch (ASO) = the modern anatomic repair; **Rastelli** for TGA + VSD + PS (LV-to-aorta baffle through the VSD + RV-to-PA conduit). Q17: it is Rastelli, NOT a univentricular/Fontan repair (answer B).
- **Eisenmenger:** a big long-standing left-to-right shunt raises the PVR above systemic, so the shunt **REVERSES to right-to-left** = cyanosis, clubbing, erythrocytosis. **INOPERABLE** (closing it is fatal), pregnancy contraindicated; treat with pulmonary vasodilators + transplant. Repair big shunts EARLY.
- **Sign-to-lesion (Q14, a match-the-column, answer E per key):** fixed split S2 = ASD; machinery murmur = PDA; holosystolic at LLSB = VSD; ejection click upper-left = pulmonary stenosis; radio-femoral delay =

coarctation; pulsatile liver + big v wave = severe TR; cyanosis + clubbing = Eisenmenger. Confirm the exact wording of option E with the course.

Key-vs-clinical conflicts to confirm with the course: Q15 (right isomerism = asplenia, key says polysplenia) and Q16 (Ebstein EXCEPT = paradoxical S2 splitting, key says ASD). Q14 exact option wording to confirm.